Project Based Assignment Report

Name – Nitin Gupta

SAP ID – 590023098

Course – Data Structures And Alogorithms

Semester – 2st

Batch – 56

School – School of Computer Science

Outputs:-

Add Student:-

```
C:\Users\gupta\OneDrive\Desktop\DSA\Assignment>a

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 1
Enter SAP ID: 590023098
Enter Name: Nitin
Enter Year: 2025
Enter Semester: 2
Enter CGPA: 8.5
Student added.

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice:
```

Adding a new student record to the linked list

Search Student

```
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 2
Enter SAP ID to search: 590023098
SAP ID: 590023098, Name: Nitin, Year: 2025, Semester: 2, CGPA: 8.50

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice:
```

Searching a student by SAP ID

Display All Students

```
SAP ID: 590023098, Name: Nitin, Year: 2025, Semester: 2, CGPA: 8.50

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 3
SAP ID: 590023098, Name: Nitin, Year: 2025, Semester: 2, CGPA: 8.50
SAP ID: 59023012, Name: Rohan, Year: 2025, Semester: 2, CGPA: 8.12
SAP ID: 590023053, Name: Ram, Year: 2025, Semester: 2, CGPA: 7.53
SAP ID: 590083224, Name: Sohan Yadav, Year: 2025, Semester: 2, CGPA: 9.25
SAP ID: 590018426, Name: Arjun Kumar, Year: 2025, Semester: 2, CGPA: 8.57
```

Displaying all student records

Delete Student

```
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 4
Enter SAP ID to delete: 59023012
Deleted.
```

Deleting a student and pushing the node onto the stack

Undo Delete

```
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 5
Restored: 59023012 Rohan
```

Restoring the deleted student using the stack (LIFO)

Sort by SAP ID

```
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 6
SAP ID: 59023012, Name: Rohan, Year: 2025, Semester: 2, CGPA: 8.12
SAP ID: 590018426, Name: Arjun Kumar, Year: 2025, Semester: 2, CGPA: 8.57
SAP ID: 590023053, Name: Ram, Year: 2025, Semester: 2, CGPA: 7.53
SAP ID: 590023098, Name: Nitin, Year: 2025, Semester: 2, CGPA: 8.50
SAP ID: 590083224, Name: Sohan Yadav, Year: 2025, Semester: 2, CGPA: 9.25
```

Linked list sorted in increasing order of SAP ID

Sort by Name

```
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 7
SAP ID: 590018426, Name: Arjun Kumar, Year: 2025, Semester: 2, CGPA: 8.57
SAP ID: 590023098, Name: Nitin, Year: 2025, Semester: 2, CGPA: 8.50
SAP ID: 590023053, Name: Ram, Year: 2025, Semester: 2, CGPA: 7.53
SAP ID: 59023012, Name: Rohan, Year: 2025, Semester: 2, CGPA: 8.12
SAP ID: 590083224, Name: Sohan Yadav, Year: 2025, Semester: 2, CGPA: 9.25
```

Linked list sorted alphabetically by name

Process Queue (FIFO)

```
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice: 8
SAP ID: 590018426, Name: Arjun Kumar, Year: 2025, Semester: 2, CGPA: 8.57
SAP ID: 590023098, Name: Nitin, Year: 2025, Semester: 2, CGPA: 8.50
SAP ID: 590023053, Name: Ram, Year: 2025, Semester: 2, CGPA: 7.53
SAP ID: 59023012, Name: Rohan, Year: 2025, Semester: 2, CGPA: 8.12
SAP ID: 590083224, Name: Sohan Yadav, Year: 2025, Semester: 2, CGPA: 9.25

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Undo Delete
6. Sort by SAP ID
7. Sort by Name
8. Process Queue (FIFO)
9. Exit
Enter choice:
```

Students processed in FIFO order using the queue